Lecture 15: Multiple Linear Regression

IEOR E4709: Statistic Analysis and Time Series
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1 Least Square Estimators and Residual: A Projection View

Recall that the least square estimator in a multiple linear regression model is given by

$$\hat{\boldsymbol{\beta}} = \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top \boldsymbol{y}.$$

Define the matrix $P_{\boldsymbol{X}} := \boldsymbol{X}\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top$ and I_n the n by n identity matrix. Both are symmetric matrices. The square matrix $P_{\boldsymbol{X}}$ is a projection matrix, i.e., $P_{\boldsymbol{X}}^2 = P_{\boldsymbol{X}}$. Moreover, $I_n - P_{\boldsymbol{X}}$ is also a projection matrix, i.e., $\left(I_n - P_{\boldsymbol{X}}\right)^2 = I_n - P_{\boldsymbol{X}}$, and $P_{\boldsymbol{X}}\left(I_n - P_{\boldsymbol{X}}\right) = 0$. Then

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}} = P_{\boldsymbol{X}}\boldsymbol{y} \tag{1}$$

and the residuals

$$\hat{\boldsymbol{\epsilon}} = \left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{y} = \left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{\epsilon}. \tag{2}$$

The last equality can also be geometrically understood from the fact that the projection of the vector $\boldsymbol{y}$ on the complement of $Span(\boldsymbol{X})$ (i.e., the span generated by the columns of the matrix $\boldsymbol{X}$) is the same as the projection of $\boldsymbol{\epsilon}$ on the complement of $Span(\boldsymbol{X})$. This is because the vector $\boldsymbol{X}\boldsymbol{\beta}$ belongs to the $Span(\boldsymbol{X})$ and hence its projection on the complement of the $Span(\boldsymbol{X})$ is zero. Formally,

$$\begin{aligned}
\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{y} &= \left(I_n - \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\right)\boldsymbol{y} \\
&= \left(I_n - \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\right)(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) \\
&= \left(I_n - \boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\right)\boldsymbol{\epsilon} \\
&= \left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{\epsilon}
\end{aligned}$$

It can be easily seen that $\hat{\boldsymbol{\beta}}$ is an unbiased estimator of $\boldsymbol{\beta}$. To see it, observe that $\mathbb{E}[\hat{\boldsymbol{\beta}}] = \left(\boldsymbol{X}^\top\boldsymbol{X}\right)^{-1}\boldsymbol{X}^\top\mathbb{E}[\boldsymbol{y}] = \left(\boldsymbol{X}^\top\boldsymbol{X}\right)^{-1}\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{\beta} = \boldsymbol{\beta}$.

It follows immediately that $\hat{\boldsymbol{\epsilon}}$ has zero mean because ϵ_i's have zero mean. Moreover, if the ϵ_i's

are Gaussian, then $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\epsilon}}$ are jointly normal. They are independent because

$$\begin{aligned} cov\left(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\epsilon}}\right) &= cov\left(\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top \boldsymbol{y}, (I_n - P_{\boldsymbol{X}})\boldsymbol{y}\right) \\ &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top cov(\boldsymbol{y})(I_n - P_{\boldsymbol{X}})^\top \\ &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top cov(\boldsymbol{\epsilon})(I_n - P_{\boldsymbol{X}})^\top \\ &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top \sigma^2 I_n (I_n - P_{\boldsymbol{X}})^\top \\ &= \sigma^2 \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \left[(I_n - P_{\boldsymbol{X}})\boldsymbol{X}\right]^\top \\ &= 0, \end{aligned}$$

where we have used that $P_{\boldsymbol{X}}\boldsymbol{X} = \boldsymbol{X}$ and that $cov(\boldsymbol{y}) = cov(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) = cov(\boldsymbol{\epsilon})$, because $\boldsymbol{X}\boldsymbol{\beta}$ is a deterministic quantity, and hence its covariance matrix is zero.

Next, let us compute the variance of the OLS estimator. We have that

$$\begin{aligned} cov\left(\hat{\boldsymbol{\beta}}\right) &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top cov\left(\boldsymbol{y}\right) \left(\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top\right)^\top \\ &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top \sigma^2 I_n \left(\left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \boldsymbol{X}^\top\right)^\top \\ &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \sigma^2 \boldsymbol{X}^\top \boldsymbol{X} \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \\ &= \left(\boldsymbol{X}^\top \boldsymbol{X}\right)^{-1} \sigma^2. \end{aligned}$$

In the third equality above, we have used that $(\boldsymbol{A}\boldsymbol{B})^\top = \boldsymbol{B}^\top \boldsymbol{A}^\top$, and that the matrix $\boldsymbol{X}^\top \boldsymbol{X}$ is symmetric.

Remark 1.1. *The OLS estimator is not only an unbiased but also a consistent estimator of $\boldsymbol{\beta}$, that is, as $n \to \infty$, the estimators $\hat{\beta}$ converge, in probability, to β. The intuition is as follows. The entries of $\boldsymbol{X}^\top \boldsymbol{X}$ are unbounded (they go to infinity) as n gets very large. For example, the $(1,1)$-th entry is n and the (j,j)-th entry is $\sum_{i=1}^n x_{ij}^2$. Therefore,*

$$\lim_{n\to\infty} (\boldsymbol{X}^\top \boldsymbol{X})^{-1} = 0,$$

and the covariance of $\hat{\boldsymbol{\beta}}$ converges to zero. This means that the distribution collapses around its expected value, namely $\boldsymbol{\beta}$. So, the limit in the probability of $\hat{\boldsymbol{\beta}}$ is given by

$$plim_{n\to\infty} \hat{\boldsymbol{\beta}} = \boldsymbol{\beta},$$

and the OLS estimator is consistent.

Denote $S_{rsdl} := \sum_{i=1}^{n} \hat{\epsilon}_i^2 = \hat{\boldsymbol{\epsilon}}^\top \hat{\boldsymbol{\epsilon}}$. Then

$$\begin{aligned}
\mathbb{E}\left[S_{rsdl}\right] &= \mathbb{E}\left[\left(\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{y}\right)^\top \left(\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{y}\right)\right] \\
&= \mathbb{E}\left[\left(\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{\epsilon}\right)^\top \left(\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{\epsilon}\right)\right] \\
&= \mathbb{E}\left[tr\left(\left(\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{\epsilon}\right)\left(\left(I_n - P_{\boldsymbol{X}}\right)\boldsymbol{\epsilon}\right)^\top\right)\right] \\
&= tr\left(\left(I_n - P_{\boldsymbol{X}}\right)\mathbb{E}\left[\boldsymbol{\epsilon}\boldsymbol{\epsilon}^\top\right]\left(I_n - P_{\boldsymbol{X}}\right)^\top\right) \\
&= tr\left(\left(I_n - P_{\boldsymbol{X}}\right)\sigma^2 I_n \left(I_n - P_{\boldsymbol{X}}\right)^\top\right) \\
&= \sigma^2\left(tr(I_n) - tr(P_{\boldsymbol{X}})\right) \\
&= (n-p)\sigma^2,
\end{aligned}$$

where in the first equality we have used the relation in (2). In the above chain of equalities, we have also used that the trace is a linear operator, $P_{\boldsymbol{X}}\left(I_n - P_{\boldsymbol{X}}\right) = 0$, and that $tr(P_{\boldsymbol{X}}) = tr(\boldsymbol{X}\left(\boldsymbol{X}^\top\boldsymbol{X}\right)^{-1}\boldsymbol{X}^\top) = tr(\boldsymbol{X}^\top\boldsymbol{X}\left(\boldsymbol{X}^\top\boldsymbol{X}\right)^{-1}) = tr(I_p) = p$ by the commutativity of the trace operator (i.e., if A is a $m \times n$ matrix and B is a $n \times m$ matrix, then $tr(AB) = tr(BA)$).

The above argument implies that an unbiased estimator of σ^2 is

$$\hat{\sigma}_{rsdl}^2 := \frac{1}{n-p} S_{rsdl}.$$

If the error terms ϵ_i's are Gaussian, the statistics $\frac{S_{rsdl}}{\sigma^2}$ follows a χ^2 distribution with $n-p$ degrees of freedom. We lose p degrees of freedom because we are estimating p parameters. We also have that, for $j = 1, \ldots, p$,

$$\frac{\hat{\beta}_j - \beta_j}{s.e(\hat{\beta}_j)}.$$

follows a Student's t-distribution with $n-p$ degrees of freedom. Above, $s.e(\hat{\beta}_j) = \sqrt{\left((\boldsymbol{X}^\top\boldsymbol{X})^{-1}\right)_{jj}}\hat{\sigma}_{rsdl}$. Using these statistics, we can then construct confidence intervals for each β_j. Moreover, we can use the above statistics to do hypothesis testing, paralleling to what we have done in the case of simple linear regression.